

AI Forecasting & Predictive Intelligence

Functional Scope, Dashboard Boards & Use Cases

TICVAI

Professional Edition

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TICVAI — Scope of Work

AI Forecasting & Predictive Intelligence — Board 1: Attendance, Demand & Revenue Forecasting

Module: AI Forecasting & Predictive Intelligence

Board: 1 of 2

Screens: 10

Board Type: Predictive Analytics / Demand Forecasting / Attendance / Revenue / Scenario Planning / Forecast Intelligence



Board Purpose

Board 1 should become TICVAI's central **predictive demand intelligence layer**.

Its purpose is to answer questions such as:

- How many visitors are expected tomorrow?
- What will attendance look like next weekend?
- Which ticket products and timeslots will experience the highest demand?
- What revenue should the venue expect?
- How does the forecast compare with current bookings?
- What happens if the venue changes price, runs a promotion, extends operating hours, or launches an event?
- How confident is TICVAI in the prediction, and what factors are driving it?
- The architecture should be:

Historical Sales + Attendance + Current Bookings + Products + Pricing + Promotions + Calendar + Operating Hours + Events + Membership + Channels + Weather/Approved External Signals + Operational History



Forecast Intelligence Engine**Attendance Forecast****Demand Forecast****Product / Channel / Timeslot Forecast****Revenue Forecast****Confidence & Uncertainty****Scenario Analysis****Forecast Recommendations / Alerts**

Board 2 and Owning TICVAI Modules

A critical principle:

Forecasting predicts and recommends. It does not automatically modify Pricing, Capacity, Workforce, Inventory or other business configurations.

Screen 1 — Forecasting Command Center

Purpose

Provide management with one central view of expected attendance, demand and revenue across venues and future periods.

Header KPIs

- Forecast Attendance Today
- Forecast Attendance Tomorrow
- Forecast Attendance Next 7 Days
- Forecast Revenue
- Current Bookings
- Expected Walk-In
- Forecast Occupancy
- Demand Index
- Forecast Confidence
- Forecast vs Actual
- Revenue Forecast Accuracy
- Active Forecast Alerts

Example**Tomorrow**

Current Bookings:

- **8,420**
- Expected Additional Sales:
- **2,180**
- Expected Walk-In:
- **1,950**
- **Forecast Attendance**
- **12,550**
- Confidence Range:
- **11,900 – 13,300**
- **Revenue Forecast**
- **AED 2.84M**
- Range:
- **AED 2.61M – AED 3.07M**
- **Forecast Trend**
- Chart:
- **Actual Historical**

vs.

- **Current Bookings**

vs.

- **Forecast**

vs.

- **Confidence Range**
- Timeline:
- Today → 30/60/90 days.
- **Demand Heatmap**
- Display:
- **Date × Time**
- Example:

Time	Mon	Tue	Wed	Thu	Fri	Sat	Sun
09–11	Low	Low	Medium	Medium	High	High	High
11–14	Medium	Medium	High	High	High	Critical	Critical
14–17	Medium	High	High	High	Critical	Critical	High
17–20	Low	Medium	Medium	High	High	High	Medium

- **Forecast Alerts**
- Examples:

HIGH DEMAND

Saturday 14:00–17:00 expected attendance 18% above normal.

REVENUE RISK

Tuesday revenue forecast 12% below target.

Filters

- Tenant
- Venue
- Date Range
- Forecast Horizon
- Product
- Channel
- Visit Date
- Customer Type
- Forecast Model
- Scenario

Actions

View Attendance | Analyze Demand | View Revenue | Run Scenario | View Alerts

Screen 2 — Forecast Configuration & Forecasting Strategy

Purpose

Configure what TICVAI should forecast, at what level of detail, for what horizon and how frequently forecasts should be refreshed.

This screen establishes the forecasting object.

Forecast Definition

Fields:

- Forecast Name
- Tenant
- Venue
- Forecast Type
- Forecast Target
- Forecast Horizon
- Time Granularity
- Refresh Frequency
- Historical Window
- Model Strategy
- Confidence Policy
- Effective Dates
- Status
- **Forecast Types**
- Support:
 - Attendance
 - Ticket Demand
 - Product Demand
 - Revenue

- • Channel Demand
 - • Timeslot Demand
 - • Event Demand
 - • Membership Demand
 - • Add-On Demand
 - • Experience Demand
 - **Forecast Horizon**
 - Examples:
 - • Intraday
 - • Next Day
 - • 7 Days
 - • 14 Days
 - • 30 Days
 - • 90 Days
 - • Seasonal
 - • Custom
 - **Granularity**
 - Configure:
 - • 15 Minutes
 - • 30 Minutes
 - • Hourly
 - • Daily
 - • Weekly
 - • Monthly
 - Not every forecast requires the same granularity.
 - For example:
 - **Venue Attendance**
 - Hourly.
 - **Annual Membership Demand**
 - Daily/Weekly.
 - **Refresh Frequency**
 - Examples:
 - • Real-Time / Event Triggered where supported
 - • Every Hour
 - • Every 4 Hours
 - • Daily
 - • Weekly
 - • Manual
 - **Forecast Hierarchy**
 - Example:
 - **Venue**
- ↓
- **Date**



- Time



- Product



Channel

TICVAI should support different forecasting dimensions without forcing every model to forecast at maximum granularity.

Model Strategy

Options conceptually:

- • Statistical
- • Machine Learning
- • AI/ML Ensemble
- • Rules/Baseline
- • Hybrid
- • Provider Model
- • Custom Tenant Model

Actual provider/model configuration belongs later to the AI Platform.

Actions

Create Forecast | Configure Strategy | Test Forecast | Activate | Clone

Screen 3 — Forecast Data & Signal Configuration

Purpose

Define which historical, current and contextual signals may contribute to each forecast.

Internal Data Sources

Potential signals:

- Sales
 - Historical Transactions
 - Booking Date
 - Visit Date
 - Quantity
 - Product
 - Channel
 - Revenue
 - Refunds
 - Cancellations
- Attendance
 - Historical Admissions
 - Entry Time
 - Exit Time

- • No-Shows
- • Re-Entries
- • Venue Occupancy
- **Ticketing**
 - • Ticket Type
 - • Product
 - • Date
 - • Timeslot
 - • Event
 - • Package
 - • Membership

Pricing

- Historical Price
- Current Price
- Price Changes

Promotions

- Promotion Period
- Discount
- Campaign / Offer
- Redemption

Capacity

- Available Capacity
- Sold Capacity
- Holds
- Closures

Calendar

- Day of Week
- Weekend
- Public Holiday
- School Holiday
- Ramadan / seasonal periods where configured
- Special Event
- Operating Hours

External Signals

Where approved:

- • Weather
- • Local Events
- • Tourism Indicators
- • School Calendar
- • Public Holidays

- • Other approved external data feeds
- **Signal Configuration**
- Each signal should include:
 - • Signal Name
 - • Source
 - • Category
 - • Historical Coverage
 - • Freshness
 - • Forecast Usage
 - • Importance / Model Use
 - • Data Quality
 - • Permission
 - • Status
- **Signal Health**
- Example:
- **Historical Attendance**
- 36 months
- Completeness:
- 99.8%
- **Weather**
- Current + Forecast
- Freshness:
- 15 min
- **Promotion History**
- 18 months
- Completeness:
- 94%
- **Missing Signal Handling**
- Configure:
 - • Ignore
 - • Use Default
 - • Use Historical Baseline
 - • Use Alternative Signal
 - • Reduce Confidence
 - • Block Forecast

Important

Missing weather data should not automatically mean:

- **Good Weather**
- It should be recorded as:

Weather Signal Unavailable

Actions

[Add Signal](#) | [View Source](#) | [Configure Usage](#) | [Test Data](#) | [View Data Quality](#)

Screen 4 — Attendance & Visitation Forecast

Purpose

Provide detailed prediction of future venue attendance.

Attendance Overview

For selected date:

- **Saturday — 19 September**
- **Current Bookings**
 - 10,820
- **Expected Future Pre-Sales**
 - 2,740
- **Expected Walk-In**
 - 3,180
- **Expected No-Shows**
 - -620
- **Forecast Attendance**
 - 16,120
- **Confidence Range**
 - 15,300 – 17,050
- **Attendance by Hour**
 - Graph:
 - 09:00 → 22:00
 - Show:
 - Forecast Entries
 - Forecast Exits
 - Expected In-Venue Population
 - Confidence Band
- **Visitor Composition**
 - Where supported:
 - Prebooked
 - Walk-In
 - Membership
 - B2B / Group
 - Reseller
 - Complimentary
 - Other configured segments
- **Attendance by Product**
 - Example:

Product	Forecast	Share
General Admission	9,200	57%

Product	Forecast	Share
Family	2,800	17%
Annual Member	1,720	11%
Group	1,400	9%
Other	1,000	6%

- **Forecast Comparison**
- **Previous Forecast**
- 15,640
- **Current Forecast**
- 16,120
- Change:
- **+3.1%**
- Reason signals:
 - Stronger booking pace
 - Weekend trend
 - Event demand
 - Weather improvement where supported

Actions

[View Hourly](#) | [View Segment](#) | [Compare Forecast](#) | [View Drivers](#) | [Send to Capacity Planning](#)

Screen 5 — Ticket, Product & Timeslot Demand Forecast

Purpose

Predict demand at the product and inventory level so TICVAI can understand **what customers are likely to buy**, not only total attendance.

Demand Forecast Table

Product	Current Sold	Forecast Demand	Remaining Opportunity	Trend
Adult Admission	4,800	7,200	+2,400	↑
Child Admission	3,100	4,650	+1,550	↑
Fast Pass	820	1,520	+700	↑
Family Meal	420	780	+360	→
Annual Pass	82	110	+28	↓

Timeslot Forecast

Example:

- **14:00–15:00**
- Capacity:
- 1,000
- Current Sold:

- 720
- Forecast Final Demand:
- **1,080**
- Expected Demand vs Capacity:
- **108%**
- Status:
- **HIGH PRESSURE**
- Forecasting should not change capacity here.

It should provide the prediction to Board 2 / Capacity Management.

Booking Curve

Show:

- **Days Before Visit**

vs.

- **Cumulative Sales**
- Compare:
 - Current Booking Curve
 - Historical Average
 - Forecast Final Demand
- **Product Relationships**
- Forecast can identify correlated demand.
- Example:

When Family Admission increases:

Family Meal demand typically increases

This is predictive intelligence, not recommendation delivery.

Unsatisfied Demand

Where enough evidence exists, estimate:

- **Potential Demand Beyond Available Inventory**
- but clearly label it as an estimate.

Actions

[View Product](#) | [View Timeslots](#) | [Compare Booking Curve](#) | [View Capacity Risk](#)

Screen 6 — Channel & Booking Pace Forecast

Purpose

Predict where and when future sales are expected to arrive.

This is particularly useful because a venue with 10,000 current bookings may still receive significant B2C, POS, reseller and walk-in volume.

Channels

Forecast:

- B2C Website
- Mobile App

- • POS
- • Flying POS
- • Kiosk
- • B2B
- • Reseller
- • OTA / API Partners where applicable
- • Call Center / Agent
- • Walk-In

- **Channel Forecast**

Channel	Current	Expected Additional	Final Forecast
B2C	6,200	1,850	8,050
App	1,420	520	1,940
B2B	1,100	380	1,480
POS/Walk-In	0	2,950	2,950

- **Booking Pace**

- Compare:

- **Current Booking Pace**

vs.

- **Historical Equivalent Days**

- Example:
- 7 days before visit:
- Current:
- 9,820 tickets
- Historical average:
- 8,400
- Difference:
- **+16.9%**

- **Pickup Forecast**

- Show expected additional bookings:

- **T-7 → T-0**

- **Pace Alerts**

- Example:
- **Saturday booking pace is 21% ahead of normal.**
- or:
- **Annual Pass renewal pace is 14% below forecast.**

- **Forecast by Lead Time**

- • Same Day
- • 1 Day
- • 2–3 Days
- • 4–7 Days
- • 8–14 Days

- 15–30 Days
- 30+ Days

Actions

[Analyze Channel](#) | [View Pickup](#) | [Compare Historical Pace](#) | [View Demand Drivers](#)

Screen 7 — Revenue & Commercial Forecast

Purpose

Forecast future revenue based on predicted sales, attendance, product mix, current pricing and other approved commercial signals.

Revenue KPIs

- Forecast Ticket Revenue
- Forecast Membership Revenue
- Forecast Add-On Revenue
- Forecast F&B Revenue where supported
- Forecast Retail Revenue where supported
- Total Forecast Revenue
- Revenue per Visitor
- Revenue vs Budget / Target where available
- Revenue Confidence Range

Example

Ticketing

AED 2.12M

Membership

AED 310K

F&B

AED 520K

Retail

AED 180K

Other

AED 95K

Total Forecast

AED 3.225M

Confidence:

- **AED 2.98M – AED 3.47M**
- **Revenue Trend**
- Compare:
 - Actual
 - Current Bookings Revenue

- • Forecast
- • Budget / Target
- • Previous Forecast
- **Revenue by Product**
- Show:
 - • Product
 - • Quantity Forecast
 - • Expected Price / Revenue Basis
 - • Forecast Revenue
 - • Contribution
- **Revenue by Channel**
- Example:
 - B2C — 48%
 - App — 14%
 - B2B — 12%
 - POS — 21%
 - Other — 5%
- **Revenue Risk**
- Example:
- **Attendance Forecast**
- +8%
- but:
- **Revenue Forecast**
- +2%
- Possible signal:

Lower expected product mix / average yield.

The system can highlight this relationship without automatically claiming a cause.

Finance Boundary

Recognized accounting revenue remains owned by Finance.

This screen provides **forecast commercial revenue**, not the accounting ledger.

Actions

View Revenue Drivers | Compare Target | View Product Mix | Run Scenario

Screen 8 — Forecast Drivers, Confidence & Explainability

Purpose

Explain why the forecast changed and how much uncertainty exists.

This is essential because management should not receive only:

- “Tomorrow attendance = 16,120.”
- They need to understand the basis and uncertainty.
- **Forecast**
- **Attendance**

- 16,120
- **Confidence Range**
- 15,300 – 17,050
- **Confidence Status**
- **High**
- **Key Drivers**
- Example:
- **Current Booking Pace**
- Positive
- +Strong
- **Saturday Historical Demand**
- Positive
- +Strong
- **Special Event**
- Positive
- +Medium
- **Weather Forecast**
- Positive
- +Medium
- **Current Price**
- Neutral
- **Recent Cancellation Rate**
- Negative
- -Medium
- **Forecast Change Explanation**
- Previous:
- 15,640
- Current:
- 16,120
- Change:
- +480
- Structured explanation:

Current booking pace increased above the historical Saturday pattern, while expected weather conditions improved. These signals contributed to an upward revision.

Uncertainty Sources

Show:

- • Limited Historical Data
- • New Product
- • New Event
- • Missing Weather
- • Abnormal Promotion
- • High Cancellation Variability
- • Unusual Booking Pattern
- **Confidence by Horizon**

- Example:
- Tomorrow:
- **92% / High**
- 7 Days:
- **84% / High**
- 30 Days:
- **71% / Medium**
- 90 Days:

58% / Lower

Do not label these as probability of being exactly correct unless that is genuinely what the metric represents.

Model Comparison

Where applicable:

- **Baseline**

vs.

- **ML**

vs.

Ensemble

Show measured historical forecast performance.

Actions

View Drivers | View Signals | Compare Models | View Historical Accuracy

Screen 9 — Forecast Scenario & What-If Simulator

Purpose

Allow management to test possible future conditions without changing production configuration.

This is one of the most important screens.

Baseline Scenario

Saturday 19 September

Attendance:

- 16,120
- Revenue:
- AED 3.225M

- **Scenario Builder**

- Change assumptions such as:
 - Ticket Price
 - Promotion
 - Operating Hours
 - Capacity
 - Event
 - Weather
 - Product Availability

- Channel Availability
- Marketing Demand Signal
- Group Booking
- Closure
- Special Day

Example Scenario

What if Adult Admission receives 10% promotion?

Simulation:

- Attendance:
- 16,120 → **17,240**
- Ticket Revenue:
- AED 2.12M → **AED 2.08M**
- F&B/Retail Forecast:
- AED 700K → **AED 762K**
- Total Revenue:
- AED 3.225M → **AED 3.292M**
- These are model estimates, not guarantees.
- **Scenario Comparison**
- Support:
- **Baseline**

vs.

- **Scenario A**

vs.

- **Scenario B**

vs.

- **Scenario C**

Example

A — 10% Promotion

B — Extend Closing +2 Hours

C — Promotion + Extended Hours

Scenario Impact

Show:

- Attendance
- Revenue
- Product Demand
- Peak Time
- Capacity Pressure
- Confidence
- Operational Implications

Important Boundary

Simulation must **not automatically change Pricing or Operating Hours.**

If management chooses a scenario:

- **Create Recommendation / Change Request**



- Owning TICVAI module.

Actions

Run Scenario | Compare Scenarios | Save Scenario | Create Recommendation

Screen 10 — Forecast Accuracy, Review & Publication Center

Purpose

Measure forecast quality over time and publish approved forecast outputs for use by other TICVAI modules.

A forecasting system should continuously answer:

- **How accurate have our forecasts actually been?**
- **Forecast vs Actual**
- Example:

Date	Forecast	Actual	Error
10 Sep	12,200	12,480	-2.2%
11 Sep	11,850	11,620	+2.0%
12 Sep	14,100	14,920	-5.5%
13 Sep	16,120	—	—

- **Accuracy Metrics**
- Where appropriate:
 - MAE
 - MAPE / WAPE
 - RMSE
 - Bias
 - Forecast Coverage
 - Confidence Interval Coverage

The UI should translate technical metrics into business-readable indicators.

Accuracy by Horizon

Next Day

96%

7-Day

91%

30-Day

84%

90-Day

72%

Only show percentage “accuracy” if calculated by an explicitly defined method.

Accuracy by Dimension

Analyze:

- • Venue
- • Product
- • Channel
- • Timeslot
- • Day Type
- • Forecast Horizon
- • Model
- • Event Type
- **Forecast Bias**
- Example:
- TICVAI consistently under-forecasts:
- **Saturday Walk-In**
- Average Bias:
- -7.4%
- Status:
- **MODEL REVIEW RECOMMENDED**
- **Forecast Version**
- Each published forecast should have:
 - • Forecast ID
 - • Version
 - • Generated Time
 - • Model Version
 - • Data Cut-Off
 - • Forecast Horizon
 - • Confidence
 - • Status
- **Publication Status**
- • Draft
- • Generated
- • Reviewed
- • Published
- • Superseded
- • Archived

Consumers

Show which TICVAI areas consume the forecast:

- • Capacity
- • Resource Management
- • Workforce
- • F&B
- • Retail
- • Inventory

- • Dynamic Pricing
- • Operations Dashboard
- • Management Reporting

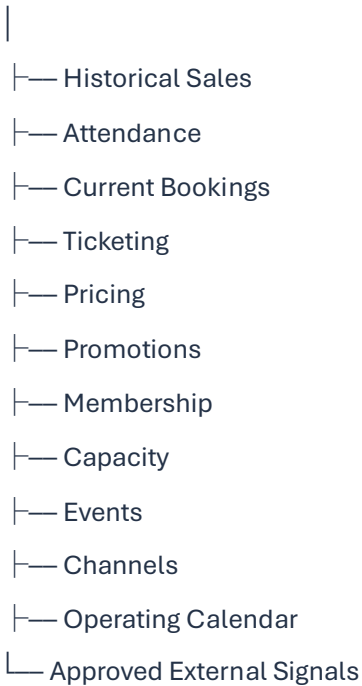
Actions

Review Forecast | Publish | Compare Actual | View Accuracy | Send to Board 2

Critical Forecasting Architecture

I recommend Soft Labs build forecasting as a **shared intelligence service**, not as separate forecasting logic inside every module.

TICVAI DATA



DATA PREPARATION / FEATURE LAYER



FORECAST INTELLIGENCE ENGINE



CONFIDENCE & EXPLAINABILITY



SCENARIO ENGINE



PUBLISHED FORECAST



Capacity	
Resource Management	
Workforce	
F&B / Retail	
Inventory	
Dynamic Pricing	
Operations	
Reporting	

Forecast Object

Every forecast should be stored as a versioned object rather than only displaying a number.

Conceptually:

- Forecast
- |
 - |— Forecast ID
 - |— Version
 - |— Tenant
 - |— Venue
- |
 - |— Forecast Type
 - |— Target
 - |— Horizon
 - |— Granularity
- |
 - |— Generated At
 - |— Data Cut-Off
- |
 - |— Model / Strategy Reference
 - |— Model Version
- |
 - |— Forecast Values
 - |— Lower Bound
 - |— Upper Bound

- |— Confidence
- |
- |— Drivers
- |— Signals Used

|— Missing Signals

|

|— Scenario Reference

|

|— Actual Result

|— Accuracy Metrics

|

|— Status

This becomes important because management may ask:

“What was the forecast we had seven days before the visit?”

The system must not overwrite that historical forecast with today's newer prediction.

Forecast Versioning

Example:

- **Saturday Attendance**
- **T-30**
- Forecast: 12,800



- **T-14**
- 14,200



- **T-7**
- 15,100



- **T-1**
- 16,120



- **Actual**
- 16,480
- TICVAI should preserve all four forecast versions.

This allows us to measure how forecast quality improves as the visit date approaches.

Forecast vs Current Bookings

Soft Labs must clearly distinguish these.

If:

- **Current Bookings = 10,000**

- and:
- **Forecast Attendance = 16,000**
- do not show:
- “AI added 6,000 customers.”
- Instead:
- **Current Confirmed Bookings**
- 10,000
- **Expected Additional Demand / Attendance Adjustments**
- +6,000
- **Forecast Final Attendance**
- 16,000

The additional figure may include future bookings, walk-ins, no-shows and other modeled components.

Walk-In Forecasting

This is especially important for attractions.

A venue may have:

- **8,000 prebooked**
- but receive:

3,000 walk-ins.

Therefore attendance forecasting cannot rely only on current ticket sales.

TICVAI should model:

Prebooked

-

Expected Future Prebookings

-

Walk-In

-

Membership Visits

-

B2B / Groups

–

Expected Cancellations

–

Expected No-Shows

=

Forecast Attendance

Hierarchical Forecast Reconciliation

Soft Labs should consider that forecasts may exist at several levels.

For example:

- **Venue Attendance**
 - 16,000
 - but product forecasts should not accidentally total:
 - 18,500
 - without explanation.
 - Where technically appropriate, TICVAI should support forecast reconciliation between:
 - **Venue**
 - ↔ **Product**
 - ↔ **Channel**
 - ↔ **Timeslot**
 - This avoids contradictory dashboards.
-

Revenue Forecasting

Revenue forecasting should combine demand prediction with the relevant commercial context.

Conceptually:

- **Predicted Quantity**
- ×
- **Expected Applicable Price / Yield**
-

Expected Product Mix

=

Forecast Revenue

But Dynamic Pricing remains separate.

Forecasting may say:

“Expected Saturday demand is high.”

Dynamic Pricing may consume that forecast and recommend a price change according to its own rules.

Forecasting itself does not modify the price.

Scenario Engine Boundary

The Scenario Engine should support hypothetical changes such as:

- What if price changes?
- But that is a **simulation input**.
- It is not Dynamic Pricing.
- Similarly:

What if capacity increases?

- does not mean Forecasting owns Capacity configuration.
 - This distinction will avoid major duplication later.
-

External Signals

External data should be pluggable rather than hard-coded.

Conceptually:

- **External Signal Adapter**
 - → Weather
 - → Public Calendar
 - → Events
 - → Tourism indicators
 - → Other approved data.

This will allow different tenants to use different signals.

External signal access must follow the AI Governance data/provider policies we just designed.

New Venue / New Product Problem

A forecasting engine needs a strategy for limited historical data.

For a newly opened venue or product, support:

- • Similar Product Baseline
- • Similar Venue Baseline
- • Category Baseline
- • Business-Provided Assumption
- • Rules-Based Baseline
- • Reduced Confidence
- • Model Transfer where technically appropriate
- The system should clearly display:

Limited Historical Data

rather than pretending the forecast has the same confidence as a venue with three years of history.

Forecast Accuracy Learning Loop

The operational loop should be:

- **Forecast**
- ↓
- **Actual**
- ↓
- **Measure Error**
- ↓
- **Identify Bias / Drift**
- ↓
- **Model Evaluation**
- ↓
- **Candidate Improvement**
- ↓

- **Validation**



Controlled Model Promotion

The model should not silently self-modify in production.

AI Governance and the future AI Platform remain responsible for controlled model lifecycle.

Board 1 → Board 2 Handover

Board 1 answers:

- **What demand is coming?**
 - Board 2 will answer:
- **What operational resources will we need because of that demand?**
 - Example:
- **Board 1**
 - Saturday forecast:
- **16,120 visitors**
 - Peak:
- **14:00–17:00**
 - Fast Pass Demand:
- **1,520**
 - F&B Demand:
- **High**



- **Board 2**
 - TICVAI determines predicted:
 - • Venue Capacity Pressure
 - • Attraction Utilization
 - • Queue Pressure
 - • Workforce Requirement
 - • POS Requirement
 - • F&B Service Demand
 - • Retail Demand
 - • Inventory Requirement
 - • Operational Resource Requirement
 - That keeps the separation very clean.

Important Boundary — Dynamic Pricing

Forecasting:

- Saturday demand expected +18%.
- Dynamic Pricing:

Based on forecast + capacity + pricing policy, recommend Adult price AED 150.

These are different responsibilities.

Important Boundary — Recommendation AI

Forecasting predicts:

- Family Meal demand = 780 units.
 - Recommendation AI decides:
 - Should Family Meal be recommended to this specific customer right now?
 - Do not combine them.
-

Important Boundary — Reporting & BI

Board 1 owns the forecasting-specific analytical experience:

- • Forecast
- • Confidence
- • Drivers
- • Accuracy
- • Scenarios
- • Forecast Versions

Enterprise Reporting can consume published forecast values for broader dashboards.

Important Boundary — Finance

Forecasting produces:

- **Forecast Revenue**
 - Finance owns:
 - • Recognized Revenue
 - • Accounting
 - • GL
 - • Reconciliation
 - • Financial Statements
 - Do not treat predicted revenue as accounting revenue.
-

Board 1 — Final Screen Map

Attendance, Demand & Revenue Forecasting

1. **Forecasting Command Center**
2. **Forecast Configuration & Forecasting Strategy**
3. **Forecast Data & Signal Configuration**
4. **Attendance & Visitation Forecast**
5. **Ticket, Product & Timeslot Demand Forecast**
6. **Channel & Booking Pace Forecast**
7. **Revenue & Commercial Forecast**
8. **Forecast Drivers, Confidence & Explainability**

9. Forecast Scenario & What-If Simulator

10. Forecast Accuracy, Review & Publication Center

This gives us a strong first forecasting board:

CONFIGURE → PREDICT → EXPLAIN → SIMULATE → MEASURE → PUBLISH

Then **Board 2 — Capacity, Resource & Workforce Forecasting** can consume these published forecasts and translate demand into operational requirements.

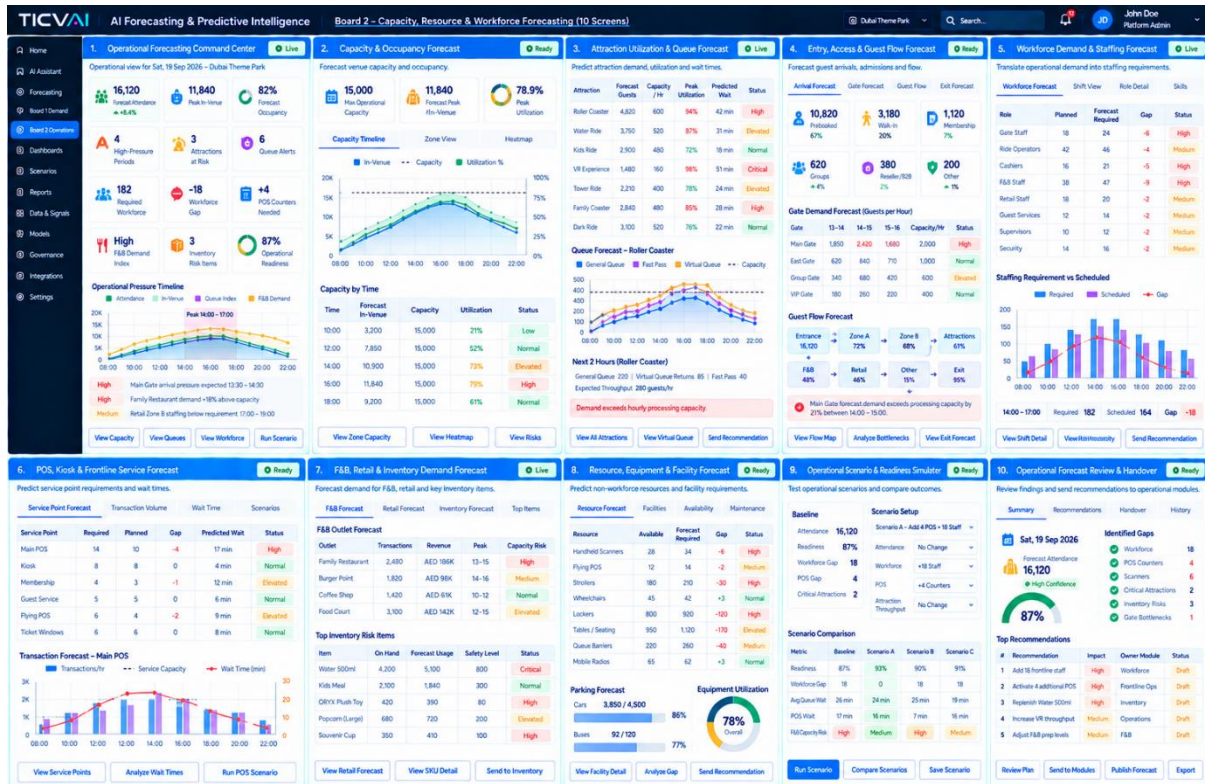
AI Forecasting & Predictive Intelligence — Board 2: Capacity, Resource & Workforce Forecasting

Module: AI Forecasting & Predictive Intelligence

Board: 2 of 2

Screens: 10

Board Type: Operational Forecasting / Capacity Pressure / Queue Demand / Workforce / F&B & Retail / Inventory / Resource Planning



Board Purpose

Board 1 answered:

- What demand is coming?
- Board 2 converts that forecast into:

What operational capacity, people, inventory and resources will TICVAI need to handle that demand effectively?

For example, Board 1 may predict:

Saturday Attendance: 16,120 visitors

Peak Period: 14:00–17:00

Fast Pass Demand: 1,520

F&B Demand: High

Walk-In: 3,180

Board 2 should translate that into operational intelligence such as:

Main Gate: High pressure expected 13:30–15:00

Attraction A: 92% expected utilization

Queue: 38-minute predicted wait

POS: 4 additional counters recommended

F&B: +22% transaction demand

Retail: +14% expected demand

Workforce: 18 additional frontline staff recommended

Inventory: selected high-volume items at stock-risk

The architecture is:

Published Board 1 Forecast

-

Capacity + Attractions + Queues + Workforce + POS + F&B + Retail + Inventory + Resources + Operating Rules



Operational Forecast Intelligence Engine



Capacity Pressure

Attraction & Queue Demand

Workforce Requirement

POS / Frontline Requirement

F&B / Retail Requirement

Inventory Requirement

Resource Requirement



Operational Gaps



Recommendations & Scenarios



Human Decision



Owning TICVAI Modules

A critical principle remains:

Forecasting predicts requirements and recommends actions. It does not directly change workforce schedules, capacity, inventory, queues, POS configuration or operating plans.

Screen 1 — Operational Forecasting Command Center

Purpose

Provide operations management with one central view of future operational pressure and predicted resource requirements.

Header KPIs

- Forecast Attendance
- Peak In-Venue Population
- Forecast Occupancy

- High-Pressure Periods
- Attractions at Risk
- Queue Pressure Alerts
- Required Workforce
- Workforce Gap
- Required POS Capacity
- F&B Demand Index
- Inventory Risk Items
- Operational Readiness Score

Example

Saturday — 19 September

Forecast Attendance:

- **16,120**
- Peak In-Venue:
- **11,840**
- Peak Period:
- **14:00–17:00**
- Forecast Occupancy:
- **82%**
- **Operational Readiness**
- **87% READY**
- Breakdown:

Area	Readiness
Venue Capacity	94%
Attractions	88%
Workforce	81%
POS	84%
F&B	86%
Retail	91%
Inventory	79%

- **Pressure Timeline**
- Show:
- 09:00 → 22:00
- with:
 - Attendance
 - Venue Occupancy
 - Gate Load
 - Attraction Pressure

- Queue Pressure
- F&B Demand
- Workforce Coverage
- **Critical Alerts**

HIGH

Main Gate arrival pressure expected 13:30–14:30.

HIGH

Family Restaurant expected demand exceeds planned service capacity by 18%.

MEDIUM

Retail Zone B staffing forecast below requirement from 17:00–19:00.

Filters

- Tenant
- Venue
- Date
- Time
- Zone
- Attraction
- Department
- Forecast Scenario

Actions

[View Capacity](#) | [View Queues](#) | [View Workforce](#) | [View F&B/Retail](#) | [Run Operational Scenario](#)

Screen 2 — Capacity & Occupancy Forecast

Purpose

Translate forecast attendance into predicted occupancy and capacity pressure across the venue.

Venue Capacity

Example:

- **Maximum Operational Capacity**
- 15,000
- **Forecast Peak In-Venue**
- 11,840
- **Forecast Utilization**
- 78.9%
- **Capacity by Time**

Time	Forecast In-Venue	Capacity	Utilization	Status
10:00	3,200	15,000	21%	Low
12:00	7,850	15,000	52%	Normal

Time	Forecast In-Venue	Capacity	Utilization	Status
14:00	10,900	15,000	73%	Elevated
16:00	11,840	15,000	79%	High
18:00	9,200	15,000	61%	Normal

- **Zone-Level Capacity**

- Break down by:

- • Entrance
- • Zone
- • Attraction
- • Experience
- • Event Space
- • F&B Area
- • Retail Area

- **Capacity Pressure Map**

- Show visual venue map / heatmap:

- **Green** — Normal

Amber — Elevated

Red — High Pressure

- **Capacity Thresholds**

- Example:

- Normal:

- <70%

- Elevated:

- 70–85%

- High:

- 85–95%

- Critical:

- 95%

- Configurable.

- **Forecast Breach**

- Example:

- **Kids Zone**

- Forecast Peak:

- 2,180

- Configured Capacity:

- 2,000

- Potential Excess Demand:

180

Forecasting should flag this but **not change the capacity**.

Actions

[View Zone](#) | [View Timeline](#) | [View Capacity Risk](#) | [Send Recommendation to Capacity](#)

Screen 3 — Attraction Utilization & Queue Forecast

Purpose

Predict demand, utilization and queue pressure for rides, attractions, experiences and service points.

Attraction Forecast

Attraction	Forecast Guests	Capacity/Hr	Peak Utilization	Predicted Wait	Status
Roller Coaster	4,820	600	94%	42 min	High
Water Ride	3,750	520	87%	31 min	Elevated
Kids Ride	2,900	480	72%	18 min	Normal
VR Experience	1,480	160	98%	51 min	Critical

Queue Forecast Timeline

Show predicted queue length:

- **09:00 → 22:00**
- Overlay:
 - General Queue
 - VIP / Fast Pass
 - Virtual Queue
 - Operational Capacity

Virtual Queue Intelligence

Where TICVAI Virtual Queue is enabled, forecast:

- Expected Virtual Queue Reservations
- Expected Return Volume
- Return-Time Concentration
- General Queue Pressure
- Priority Queue Pressure

This is particularly important because virtual queue customers return at scheduled periods and create predictable arrival waves.

Example

VR Experience

14:00–15:00

General Queue:

- 220 guests
- Virtual Queue Returns:
 - 85 guests
 - VIP / Fast Pass:
 - 40 guests
- Expected Throughput:
 - 280 guests
- **Result**
 - **Demand exceeds hourly processing capacity.**

- **Operational Recommendation**

- Examples:
 - Increase operating capacity where possible
 - Modify queue allocation
 - Open additional loading position
 - Review virtual queue release volume
 - Temporarily reduce future virtual queue inventory
- These remain recommendations to the owning modules.

Actions

[View Attraction](#) | [Analyze Queue](#) | [View Virtual Queue](#) | [Send Operational Recommendation](#)

Screen 4 — Entry, Access & Guest Flow Forecast

Purpose

Predict guest arrival, admission, exit and movement pressure across gates and venue zones.

Arrival Forecast

Break down expected arrivals by:

- Prebooked
- Walk-In
- Membership
- Groups
- B2B
- Reseller
- VIP
- Other configured categories

- **Gate Demand**

Gate	13-14	14-15	15-16	Capacity/Hr	Risk
Main Gate	1,850	2,420	1,680	2,000	High
East Gate	620	840	710	1,000	Normal
Group Gate	340	680	420	600	Elevated

- **Entry Processing Forecast**

- Forecast:
 - Scans per Minute
 - Guests per Hour
 - Group Arrivals
 - Credential Mix
 - Manual Validation Requirement
 - Re-Entry Volume

- **Guest Flow**

- Predict:

- **Entrance**
- → **Zone A**
- → **Zone B**
- → **Attractions**
- → **F&B**
- → **Retail**
- → **Exit**
- where sufficient operational data exists.
- **Bottleneck Alert**
- Example:

Main Gate forecast demand between 14:00–15:00 exceeds expected processing capacity by 21%.

Exit Forecast

Forecast departures by hour to support:

- • Exit Gate Staffing
- • Transport
- • Parking
- • Guest Flow
- • Security

Actions

[View Gate Forecast](#) | [View Guest Flow](#) | [Analyze Bottleneck](#) | [Send to Access Operations](#)

Screen 5 — Workforce Demand & Staffing Forecast

Purpose

Translate predicted operational demand into required staffing levels.

Workforce Requirement

Forecast by:

- • Department
- • Role
- • Venue
- • Zone
- • Attraction
- • Service Point
- • Shift
- • Hour

Example

Role	Planned	Forecast Required	Gap	Status
Gate Staff	18	24	-6	High
Ride Operators	42	46	-4	Medium

Role	Planned	Forecast Required	Gap	Status
Cashiers	16	21	-5	High
F&B Staff	38	47	-9	High
Retail Staff	18	20	-2	Medium
Guest Services	12	14	-2	Medium

Staffing Timeline

Show:

- **Required Staff**

vs.

- **Currently Scheduled**
- hour by hour.
- **Staffing Drivers**
- Example:
- **Gate Staff Requirement**
- Driven by:
 - Forecast arrival volume
 - Gate throughput
 - Group arrivals
 - Credential mix
 - Manual validation rate

Role Productivity

Where configured, use operational standards such as:

- **Cashier**
- Expected transactions/hour
- **Ride Operator**
- Required positions per operating mode
- **F&B Staff**
- Transactions/hour

Skills & Certification

Forecast should respect requirements such as:

- Certified Ride Operator
- Lifeguard
- Security
- Cashier
- Food Handling
- First Aid
- Supervisor
- It should not simply say:
 - “Need 10 more employees”
 - when specific qualified roles are required.

- **Workforce Gap**
- Example:
- 14:00–17:00:
- **Required: 182**
- **Scheduled: 164**
- **Gap: 18**

Actions

[View Role Detail](#) | [View Shift Gap](#) | [Create Staffing Recommendation](#) | [Send to Resource Management](#)

Screen 6 — POS, Kiosk & Frontline Service Forecast

Purpose

Predict the number of active selling/service points required to handle expected transaction volume and maintain acceptable service levels.

Service Points

Forecast:

- POS Counters
- Flying POS
- Ticket Windows
- Kiosks
- Guest Service Desks
- Membership Counters
- F&B POS
- Retail POS

Example

Main Ticketing Area

Forecast Transactions:

- 1,480/hour
- Average Service Rate:
- 110 transactions/hour/counter
- **Recommended Active Counters**
- **14**
- Currently Planned:
- **10**
- Gap:
- **4**
- **Service Forecast**

Area	Required	Planned	Gap	Predicted Wait
Main POS	14	10	-4	17 min
Kiosk	8	8	0	4 min

Area	Required	Planned	Gap	Predicted Wait
Membership	4	3	-1	12 min
Guest Service	5	5	0	6 min

- **Flying POS Recommendation**

- Example:

Deploy 2 Flying POS terminals near Main Gate between 13:30–15:30.

This is a recommendation.

Frontline/POS Operations controls actual deployment.

Failure Scenario

Forecast can consider:

- **1 POS unavailable**
- and show expected impact.

Actions

[View Service Point](#) | [Analyze Wait](#) | [Create POS Recommendation](#) | [Run Scenario](#)

Screen 7 — F&B, Retail & Inventory Demand Forecast

Purpose

Translate visitor forecasts into expected F&B, retail and stock demand.

This screen should forecast operational requirements without replacing the detailed F&B/Retail/Inventory modules already designed.

F&B Forecast

Example:

Outlet	Transactions	Revenue	Peak	Capacity Risk
Family Restaurant	2,480	AED 186K	13–15	High
Burger Point	1,820	AED 98K	14–16	Medium
Coffee Shop	1,420	AED 61K	10–12	Normal

- **Retail Forecast**

- Forecast:
 - Transactions
 - Units
 - Revenue
 - Category Demand
 - Peak Time

- **Product / SKU Demand**

- Where supported:

Item	On Hand	Forecast Usage/Sales	Safety Level	Risk
Water 500ml	4,200	5,100	800	Critical
Kids Meal	2,100	1,840	300	Normal
ORYX Plush Toy	420	390	80	High

- **Stock Risk**
- Calculate:
- **Available Stock**

vs.

- **Forecast Consumption**

vs.

- **Safety Stock**
- **Recommendation**
- Example:

Water 500ml forecast requirement exceeds available inventory by 900 units before safety stock.

Recommended:

- **Replenishment Review**
- Not:

Automatically create PO

unless the Inventory/Procurement workflow explicitly allows it.

F&B Preparation Forecast

Where relevant:

- • Meal Volume
- • Ingredient Demand
- • Preparation Load
- • Service Demand

Actions

[View Outlet](#) | [View SKU Forecast](#) | [View Stock Risk](#) | [Send to Inventory Planning](#)

Screen 8 — Resource, Equipment & Facility Requirement Forecast

Purpose

Predict non-workforce operational resources required to support expected visitor demand.

Resource Types

Examples:

- • Handheld Scanners
- • Mobile POS
- • Turnstile Lanes
- • Kiosks
- • Wheelchairs

- Strollers
- Lockers
- Rental Equipment
- Tables / Seating
- Parking Capacity
- Event Equipment
- Temporary Counters
- Guest Service Equipment

• Resource Forecast

Resource	Available	Forecast Required	Gap	Status
Handheld Scanners	28	34	-6	High
Flying POS	12	14	-2	Medium
Strollers	180	210	-30	High
Wheelchairs	45	42	+3	Normal
Lockers	800	920	-120	High

• Facility Forecast

- Predict demand for:
 - Parking
 - Seating
 - Lockers
 - Rental Counters
 - Rest Areas
 - Service Desks
- where data supports forecasting.

• Availability

- Resource Management remains the source of:
 - Resource Inventory
 - Availability
 - Maintenance Status
 - Reservation
- Forecasting consumes that information.

Example

Handheld Scanners

Forecast Required:

- 34
- Inventory:
- 38
- Under Maintenance:
- 10
- Operationally Available:

- 28
- **Gap**
- 6

Actions

[View Resource](#) | [Analyze Gap](#) | [Create Requirement Recommendation](#) | [Send to Resource Management](#)

Screen 9 — Operational Scenario & Readiness Simulator

Purpose

Allow management to test operational alternatives before changing real configurations.

Baseline

Attendance

16,120

Operational Readiness

87%

Workforce Gap

18

Critical Attractions

2

POS Gap

4

Inventory Risks

3

Scenario A

Add 18 Staff

Results:

- Workforce Readiness:
 - 81% → **98%**
- Overall Readiness:
 - 87% → **93%**
- **Scenario B**
 - **Open 4 Additional POS**
 - Expected Ticketing Wait:
 - 17 min → **7 min**
 - Overall Readiness:
 - 87% → **90%**
- **Scenario C**
 - **Increase Attraction Throughput +10%**
 - Expected Queue:
 - 42 min → **31 min**

- **Scenario D**
- **Attendance +15%**
- Stress test:
- Forecast Attendance:
- 18,538
- Critical Capacity Areas:
- 2 → 7
- Workforce Gap:
- 18 → 42
- **Adjustable Variables**
- Support hypothetical:
 - Attendance
 - Operating Hours
 - Capacity
 - Attraction Throughput
 - Workforce
 - POS
 - Gate Availability
 - F&B Service Capacity
 - Inventory
 - Resource Availability
 - Attraction Closure
 - Weather Scenario
 - Group Arrival
- **Scenario Comparison**

Metric	Baseline	A	B	C
Readiness	87%	93%	90%	91%
Workforce Gap	18	0	18	18
Avg Queue	26m	24m	25m	19m
POS Wait	17m	16m	7m	16m

Important

Simulation does not execute production changes.

Actions

Run Scenario | Compare | Save | Create Operational Plan Recommendation

Screen 10 — Operational Forecast Review, Recommendations & Handover

Purpose

Consolidate forecast findings into a controlled operational readiness plan and hand recommendations to the appropriate TICVAI modules.

This is the final screen of the Forecasting module.

Forecast Summary

Date

Saturday — 19 September

Forecast Attendance

16,120

Confidence

High

Operational Status

Readiness: 87%

Identified Gaps

- Workforce: 18
- POS Counters: 4
- Handheld Scanners: 6
- High-Pressure Attractions: 2
- Inventory Risks: 3
- Gate Bottlenecks: 1

AI Operational Recommendations

Example:

- **Recommendation 1**

- **Workforce**

- Add:

- • 6 Gate Staff
- • 4 Ride Operators
- • 5 Cashiers
- • 3 F&B Staff

- Period:

- 13:00–17:00

- Confidence:

High

Recommendation 2

POS

Activate 4 additional selling points.

Period:

- 13:30–15:30
- Expected effect:

Reduce forecast wait from 17 minutes to approximately 7 minutes.

Recommendation 3

Inventory

Review replenishment of:

- Water 500ml
- Forecast shortage:
- 900 units before safety stock.
- **Recommendation Status**
- Each recommendation:
 - • Draft
 - • Reviewed
 - • Accepted
 - • Rejected
 - • Sent to Owning Module
 - • Actioned
 - • Superseded
- **Handover**
- Examples:
- **Capacity Recommendation**
- → Capacity Management
- **Virtual Queue Recommendation**
- → Virtual Queue
- **Workforce Recommendation**
- → Resource / Workforce Management
- **POS Recommendation**
- → Frontline Operations
- **Inventory Recommendation**
- → Inventory & Procurement
- **F&B Recommendation**
- → F&B
- **Human Decision**
- Operations Manager can:
- **Accept**
- **Reject**
- **Modify**
- **Send for Planning**
- **Forecast Version Binding**
- The operational plan must reference:
- **Forecast Version**
- FRC-20260919-v7
- If the forecast materially changes:

Operational Plan Review Recommended

Actions

Review Recommendations | Accept Selected | Send to Modules | Publish Operational Forecast

Critical Board 1 → Board 2 Architecture

The relationship between the two boards should be explicit:

- BOARD 1
- ATTENDANCE, DEMAND & REVENUE FORECASTING

- |
- |— Attendance
- |— Arrival Pattern
- |— Product Demand
- |— Timeslot Demand
- |— Channel Demand
- |— Walk-In
- |— F&B / Retail Demand
- |— Revenue



- PUBLISHED FORECAST



- BOARD 2
- OPERATIONAL FORECAST INTELLIGENCE



- |
- | Venue Capacity Pressure |
- | Attraction Utilization |
- | Queue Demand |
- | Entry / Guest Flow |
- | Workforce Requirement |
- | POS / Service Point Requirement |
- | F&B / Retail Requirement |
- | Inventory Requirement |
- | Equipment / Resource Requirement |



GAP ANALYSIS



OPERATIONAL SCENARIOS



RECOMMENDATIONS



HUMAN REVIEW



OWNING MODULES

Operational Requirement Object

I recommend Soft Labs create a standardized forecast requirement object.

Conceptually:



This allows all downstream modules to consume forecasts consistently.

Current Plan vs Forecast Requirement

This distinction is extremely important.

TICVAI should show:

- **Currently Planned**
 - versus:
 - **Forecast Required**
 - versus:
 - **Gap**
 - For example:
 - **Gate Staff**
 - Planned:
 - 18
 - Forecast Required:
 - 24
 - Gap:
 - **-6**
 - Forecasting should not silently change:
 - **Scheduled Staff = 24**
 - The Operations/Workforce team must decide what to do.
-

Demand Is Not the Same as Capacity

Another important distinction for Soft Labs:

- If forecast demand is:
- **1,080 guests**
- and timeslot capacity is:

1,000

this does not mean forecast attendance automatically becomes 1,000.

Instead:

- **Forecast Demand**
 - 1,080
 - **Available Capacity**
 - 1,000
 - **Potential Excess / Unsatisfied Demand**
 - 80
 - This difference is commercially useful.
-

Queue Forecasting Architecture

Queue prediction should conceptually consider:

- **Arrival Rate**
- versus:
- **Processing / Throughput Rate**
- plus:
 - • Operating Capacity
 - • Downtime

- • VIP/Fast Pass Allocation
- • Virtual Queue Returns
- • Group Arrivals
- • Guest Behavior
- • Historical Service Times
- This is particularly important for TICVAI because we already designed:
 - • General Queue
 - • VIP / Priority
 - • Virtual Queue

Forecasting should understand all three as competing demand streams.

Virtual Queue Forecasting

For Virtual Queue, Board 2 should forecast not only:

- How many people will reserve?
- but:
- **When will they return?**
- Example:
- 14:00:

200 general queue guests

-

90 virtual queue returns

-

40 VIP/Fast Pass

=

330 guests requiring service

This is what operations actually needs to plan.

Workforce Forecasting

Workforce requirement should not be:

Visitors ÷ arbitrary number.

It should use configurable productivity and operational rules.

For example:

- **Gate**
- Forecast:
- 2,400 guests/hour
- Configured processing rate:
- 120 guests/hour/operator
- Base requirement:

20 positions

-

Supervision

-

Break Coverage

-

Operational Buffer

=

Forecast Workforce Requirement

This allows different tenants to configure their own operational standards.

Workforce Scheduling Boundary

Board 2 predicts:

- 24 Gate Staff required.
- Resource/Workforce Management decides:
- Which 24 employees work.
- Forecasting should not own:
 - Employee Rosters
 - Shift Assignment
 - Leave
 - Payroll
 - Attendance
 - Employee Contracts

Skills Matter

If an attraction requires:

- **4 certified operators**
- TICVAI cannot solve the gap by recommending:
- 4 generic employees.
- Forecast requirements should therefore support:

Role

-

Skill

-

Certification

-

Location

-

Time

F&B Forecasting Boundary

Board 2 may predict:

Family Restaurant requires approximately 2,480 transactions.

The F&B module owns:

- • Menu
- • Kitchen
- • Recipes
- • Orders
- • POS
- • Outlet Configuration

Forecasting only predicts demand and required operational capacity.

Inventory Boundary

Board 2 can identify:

- Forecast stock shortage.
- Inventory & Procurement decides:
 - • Replenishment
 - • Transfer
 - • Purchase Order
 - • Supplier
 - • Receiving

This prevents duplication of the Inventory boards already completed.

Resource Management Boundary

Board 2 predicts:

- 34 scanners required.
- Resource Management knows:
 - • 38 owned
 - • 10 under maintenance
 - • 28 available
- Board 2 therefore calculates:
- **Required 34**

vs.

- **Operationally Available 28**
 - =
 - **Gap 6**
 - It then recommends action.
-

Operational Readiness Score

If we use the readiness score, it must be deterministic and transparent.

Example configurable calculation:

- • Capacity Readiness — 20%
- • Attraction/Queue Readiness — 20%
- • Workforce — 20%
- • POS/Frontline — 10%
- • F&B/Retail — 10%
- • Inventory — 10%
- • Resources — 10%
- Result:

87% Readiness

Management should be able to click the score and see exactly why it is 87%.

An LLM should not simply invent the score.

Forecast Confidence

Board 2 should inherit Board 1 uncertainty.

If Board 1 says:

- **Attendance Forecast Confidence = Low**
 - Board 2 should not present:
 - “Exactly 24 cashiers required.”
 - Instead:
 - **Expected Requirement: 22–26 cashiers**
 - Recommended Planning Level:
 - 24
 - Confidence:
 - Medium
 - This is much more operationally realistic.
-

Forecast Change Impact

Operational forecasts must remain linked to their source forecast version.

Example:

- At 08:00:
- Attendance forecast:
- 14,800
- Workforce requirement:
- 162
- At 12:00:
- Attendance forecast changes:
- 16,120
- Workforce requirement:
- 182
- TICVAI should show:

Forecast increased by 8.9%. Operational plan may require review.

It should not silently replace the old plan without notifying operations.

AI Recommendations

AI can add value by saying:

“The largest operational gap is forecast between 13:30 and 16:00. Adding four POS counters and six gate staff would address the two highest predicted bottlenecks.”

But the recommendation should be based on structured forecast data.

AI is helping interpret the operational forecast—not inventing operational facts.

Governance Integration

Because we already completed AI Governance:

- **Low-Risk**
- Forecast generation:
- **Automatic**
- **Medium-Risk**
- Operational recommendation:

Automatic recommendation / human review

Higher-Risk

If TICVAI later allows AI to initiate operational changes:

- **Governance Policy**



- **Approval**



Owning Module

Forecasting itself should remain advisory by default.

Board 2 — Final Screen Map

Capacity, Resource & Workforce Forecasting

1. **Operational Forecasting Command Center**
2. **Capacity & Occupancy Forecast**
3. **Attraction Utilization & Queue Forecast**
4. **Entry, Access & Guest Flow Forecast**
5. **Workforce Demand & Staffing Forecast**
6. **POS, Kiosk & Frontline Service Forecast**
7. **F&B, Retail & Inventory Demand Forecast**
8. **Resource, Equipment & Facility Requirement Forecast**
9. **Operational Scenario & Readiness Simulator**
10. **Operational Forecast Review, Recommendations & Handover**

With this, the **AI Forecasting & Predictive Intelligence** module is fully scoped at 2 boards / 20 screens:

Board 1 — What demand is coming?**Attendance → Demand → Revenue → Confidence → Scenario****Board 2 — What will we need to handle it?****Capacity → Queue → Guest Flow → Workforce → POS → F&B/Retail → Inventory → Resources → Operational Readiness**

The next design step is **Design Board 2**. After that, Forecasting is complete and we can move to the **AI Fraud & Risk Intelligence module**.